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NAAC GRADE A++

Information and Communication Technology

Artificial Intelligence
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CHAPTER-6

Natural Language Processing

Topics to be Covered

- Introduction
- Syntactic Processing
- Semantic Analysis
- Discourse and Pragmatic Processing
- Spell Checking

Introduction

- Natural Language Processing (NLP) is an interdisciplinary field that combines computer science, linguistics, and artificial intelligence to enable machines to understand, interpret, and generate human language.

Introduction

- Human language is incredibly complex and diverse. It encompasses syntax (structure), semantics (meaning), pragmatics (context), and even the socio-linguistic elements that influence communication.
- NLP aims to bridge the gap between human communication and computer understanding, facilitating smoother and more intuitive interactions.

Introduction

- The ultimate goal of NLP is to help computers understand human language (as good as we humans understand).
- Speech recognition —the translation of spoken language into text.
- Natural language understanding —a computer’s ability to understand language.
- Natural language generation —the generation of natural language by a computer.

Components of Natural Language Processing (NLP)

Lexical Analysis
↓
Syntactic Analysis
↓
Semantic Analysis
↓
Semantic Analysis
↓
Semantic Analysis

Components of Natural Language Processing (NLP)

- Lexical Analysis: With lexical analysis, we divide a whole chunk of text into paragraphs, sentences, and words. It involves identifying and analyzing words' structure.
- Syntactic Analysis: It involves the analysis of words in a sentence for grammar and arranging words in a manner that shows the relationship among the words. For instance, the sentence "The shop goes to the house" does not pass.

Components of Natural Language Processing (NLP)

- Semantic Analysis: It draws the exact meaning for the words, and it analyzes the text meaningfulness. Sentences such as “hot ice-cream” do not pass.
- Disclosure Integration: It takes into account the context of the text. It considers the meaning of the sentence before it ends. For example: “He works at Google.” In this sentence, “he” must be referenced in the sentence before it.

Components of Natural Language Processing (NLP)

- Pragmatic Analysis: It deals with overall communication and interpretation of language. It deals with deriving meaningful use of language in various situations.

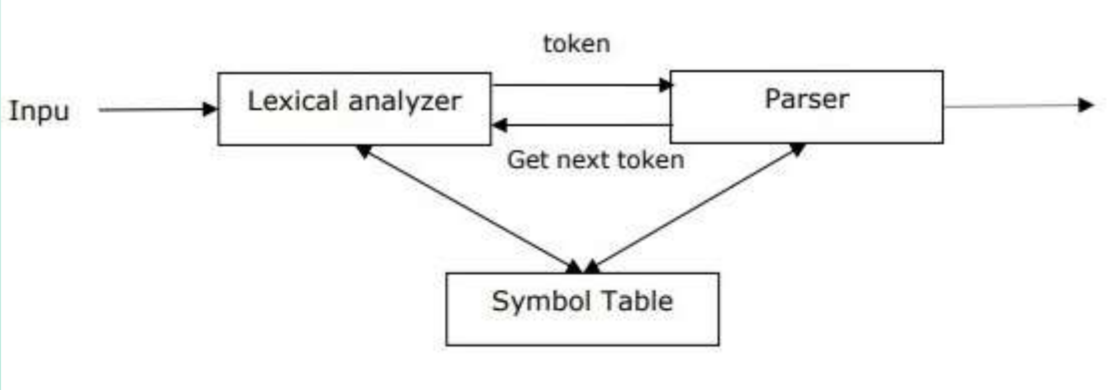
Syntactic Processing

- Syntactic analysis or parsing is the third phase of NLP.
- The purpose of this phase is to draw exact meaning, or you can say dictionary meaning from the text.
- Syntax analysis checks the text for meaningfulness comparing to the rules of formal grammar.
- For example, the sentence like “hot ice-cream” would be rejected by semantic analyzer.

Syntactic Processing

- In this sense, syntactic analysis or parsing may be defined as the process of analyzing the strings of symbols in natural language conforming to the rules of formal grammar.
- The origin of the word ‘parsing’ is from Latin word ‘pars’ which means ‘part’.

Syntactic Processing



- It may be defined as the software component designed for taking input data (text) and giving structural representation of the input after checking for correct syntax as per formal grammar.

Syntactic Processing

- It also builds a data structure generally in the form of parse tree or abstract syntax tree or other hierarchical structure.
- The main roles of the parse includes:
  - To report any syntax error.
  - To recover from commonly occurring error so that the processing of the remainder of program can be continued.
  - To create parse tree.
  - To create symbol table.
  - To produce intermediate representations (IR).

Syntactic Processing

- Types of Parsing:
  - Top-down parsing:
    - In this kind of parsing, the parser starts constructing the parse tree from the start symbol and then tries to transform the start symbol to the input. The most common form of top-down parsing uses recursive procedure to process the input. The main disadvantage of recursive descent parsing is backtracking.

Syntactic Processing

- Bottom-up parsing:
  - In this kind of parsing, the parser starts with the input symbol and tries to construct the parser tree up to the start symbol.

- Derivation:
  - In order to get the input string, we need a sequence of production rules. Derivation is a set of production rules.
  - During parsing, we need to decide the non-terminal, which is to be replaced along with deciding the production rule with the help of which the non-terminal will be replaced.

Syntactic Processing

- Types of Derivation:
  - Left-most Derivation:
    - In the left-most derivation, the sentential form of an input is scanned and replaced from the left to the right. The sentential form in this case is called the left-sentential form.
  - Right-most Derivation:
    - In the left-most derivation, the sentential form of an input is scanned and replaced from right to left. The sentential form in this case is called the right-sentential form.

Syntactic Processing

- Parse Tree:
  - It may be defined as the graphical depiction of a derivation.
  - The start symbol of derivation serves as the root of the parse tree.
  - In every parse tree, the leaf nodes are terminals and interior nodes are non-terminals.
  - A property of parse tree is that in-order traversal will produce the original input string.

Discourse Processing

- Discourse in Natural Language Processing (NLP) refers to the study and analysis of how sequences of sentences relate to each other to form coherent and meaningful communication.
- Unlike the analysis of isolated sentences, discourse analysis looks at the flow and structure of larger pieces of text or conversation.
- It is essential for understanding context, maintaining coherence, and enabling more natural and effective human-computer interactions.

Discourse Processing

- Discourse encompasses written or spoken communication longer than a single sentence. It involves:
  - **Coherence:** Ensuring that the text or speech logically flows from one sentence to the next.
  - **Context:** Understanding the broader situation or background information that influences the meaning of the text.
  - **Structure:** Recognizing the organization and patterns in communication, such as narrative structure, argumentation, and dialogue.

Discourse Processing

- **Discourse segmentation:**
  - Discourse segmentation can be defined as the process of determining the types of structures that will be used in large amounts of discourse.
  - Implementing discourse segmentation is difficult, but it is critical for applications such as information retrieval, text summarization, and information extraction.

Discourse Processing

- Discourse Integration is dependent on the sentences that come before it and also invokes the meaning of the sentences that come after it.
- Unsupervised Discourse Segmentation and Supervised Discourse Segmentation are popularly used algorithms for discourse segmentation.

Pragmatics Processing

- Pragmatics is a branch of linguistics dealing with language use in context.
- Pragmatics is the study of how linguistic properties and contextual factors interact in the interpretation of utterances, enabling hearers to bridge the gap between sentence meaning and speaker's meaning.

Pragmatics Processing

- Pragmatics is the study of how context influences the interpretation of meaning in communication.
- It goes beyond the literal meanings of words and sentences to consider factors such as:
  - **Speaker Intentions:** What the speaker intends to convey.
  - **Contextual Information:** The situational context in which communication occurs.

Pragmatics Processing

- **Social and Cultural Norms:** The norms and conventions that influence how language is used and understood.
- **Implicature:** The implied meaning that is not explicitly stated.

Pragmatics Processing

- Applications of Pragmatic Processing:
  - **Dialogue Systems:** Enhancing virtual assistants and chatbots to understand and respond appropriately to user intentions and context.
  - **Machine Translation:** Improving the quality of translations by considering the pragmatic context to ensure the intended meaning is preserved.

Pragmatics Processing

- **Sentiment Analysis:** Better understanding the sentiment behind text by considering context and implied meanings.
- **Human-Robot Interaction:** Enabling robots to interpret and respond to human actions and speech in contextually appropriate ways.

Pragmatics Processing

- Challenges in Pragmatic Processing:
  - Context Dependency
  - Ambiguity
  - Cultural and Social Norms
  - Computational Complexity

Spell Checking

- Spell checking is an essential component of Natural Language Processing (NLP) that focuses on identifying and correcting spelling errors in text.
- It enhances the accuracy and readability of written communication, making it a fundamental feature in word processors, search engines, and various other applications.
- Spell checking involves detecting misspelled words and suggesting possible corrections.

Spell Checking

- Types of Spelling Errors:
  - **Typographical Errors (Typos):** Mistakes made while typing, such as "teh" instead of "the."
  - **Orthographic Errors:** Errors related to the correct spelling of words, such as "recieve" instead of "receive."
  - **Contextual Errors:** Errors that are context-dependent, where a word is spelled correctly but used incorrectly, such as "their" instead of "there."

Spell Checking

- Components of Spell Checking:
  - **Error Detection:** Error detection is the process of identifying words that are likely misspelled.
  - **Error Correction:** Once an error is detected, the next step is to suggest possible corrections. This involves generating a list of candidate words that are similar to the misspelled word and selecting the most likely correction.

Spell Checking

- **Contextual Analysis:** For contextual errors, spell checkers must consider the surrounding text to make accurate corrections. This involves understanding the context in which a word is used.

Spell Checking

- Applications of Spell Checking:
  - Word Processors
  - Search Engines
  - Text Input Interfaces
  - Language Learning Tools

Spell Checking

- Challenges in Spell Checking:
  - Homophones
  - Proper Names and Technical Terms
  - Multilingual Support
  - Typographical Variations

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